

## Unit 6

# ELECTROCHEMISTRY

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24 items · 43 marks

### **Syllabus sections in this unit**

6.1 Redox processes: electron transfer and changes in oxidation number (oxidation state)

## Unit 6: Electrochemistry

### 1. M/J 26/P21/Q3(d)

Context

- 3 Separate samples of NaCl, NaBr and NaI each react with concentrated H<sub>2</sub>SO<sub>4</sub>. Table 3.1 shows the observations for each reaction.

Table 3.1

| reaction | compound | observations                                                                                                                                                                                                                                  |
|----------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | NaCl     | <ul style="list-style-type: none"> <li>effervescence</li> <li>steamy fumes that turn blue litmus paper red</li> </ul>                                                                                                                         |
| 2        | NaBr     | <ul style="list-style-type: none"> <li>effervescence</li> <li>steamy fumes initially that turn blue litmus paper red</li> <li>brown vapour</li> <li>pungent smell of SO<sub>2</sub></li> </ul>                                                |
| 3        | NaI      | <ul style="list-style-type: none"> <li>effervescence</li> <li>steamy fumes initially that turn blue litmus paper red</li> <li>purple vapour</li> <li>pungent smell of SO<sub>2</sub> and then a rotten egg smell of H<sub>2</sub>S</li> </ul> |

- (d) Complete Table 3.2.

Table 3.2

| species                        | oxidation number of sulfur |
|--------------------------------|----------------------------|
| H <sub>2</sub> SO <sub>4</sub> |                            |
| SO <sub>2</sub>                |                            |
| H <sub>2</sub> S               |                            |

[2]

### 2. M/J 26/P21/Q4(a)(ii)

Context: part (a) (introduction)

- 4 (a) Compound **Z** is oxidised to produce organic compound **W**, C<sub>5</sub>H<sub>6</sub>O<sub>7</sub>, by heating under reflux with an excess of acidified Cr<sub>2</sub>O<sub>7</sub><sup>2-</sup>(aq).

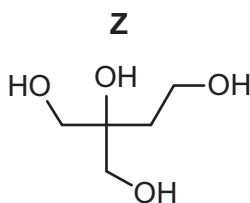
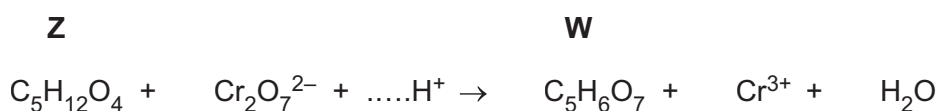


Figure 4.1

- (ii) The complete oxidation of 0.025 mol of **Z** requires 0.050 mol of Cr<sub>2</sub>O<sub>7</sub><sup>2-</sup>(aq). **W** is produced in this reaction.

Balance the equation for this reaction.

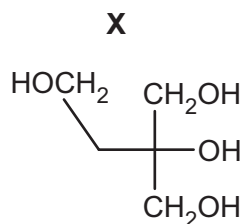


[1]

**3. M/J 26/P22/Q4(c)**

also Unit 17 (2 marks); this unit 2 marks

Context: part (b) (introduction)

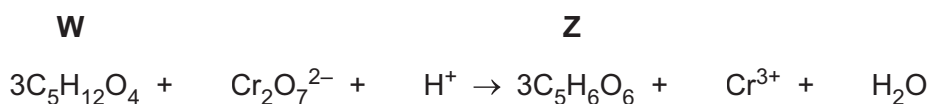
**(b)** The structure of compound **X** is shown in Figure 4.2.**Figure 4.2**

The molecular formula of **X** is  $\text{C}_5\text{H}_{12}\text{O}_4$ . Complete oxidation of 0.255 g of **X** occurs when it reacts with exactly  $10.0 \text{ cm}^3$  of  $0.500 \text{ mol dm}^{-3}$  acidified  $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$ .

**(c) (i)** Compounds **X** and **W** are a pair of structural isomers.

0.0012 mol of **W** is completely oxidised when it is heated under reflux with exactly 0.0020 mol of acidified  $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$  to produce compound **Z**.

Complete the equation for this reaction.



[2]

**(ii)** Each molecule of **W** contains four hydroxyl groups. Only two of these are primary hydroxyl groups.

Table 4.1 shows the observations when separate samples of **Z** are added to each reagent.

**Table 4.1**

| reagent                               | observations with <b>Z</b> |
|---------------------------------------|----------------------------|
| 2,4-dinitrophenylhydrazine (2,4-DNPH) | orange precipitate         |
| Fehling's reagent                     | none                       |

Use the information in **4(c)** to suggest the structure of **W**.

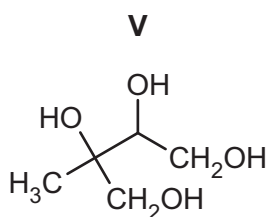


[2]

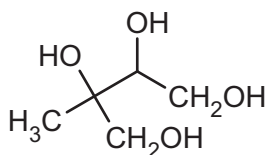
**4. M/J 26/P23/Q4(d)(ii)**

Context

- 4** The structure of compound **V** is shown in Figure 4.1.

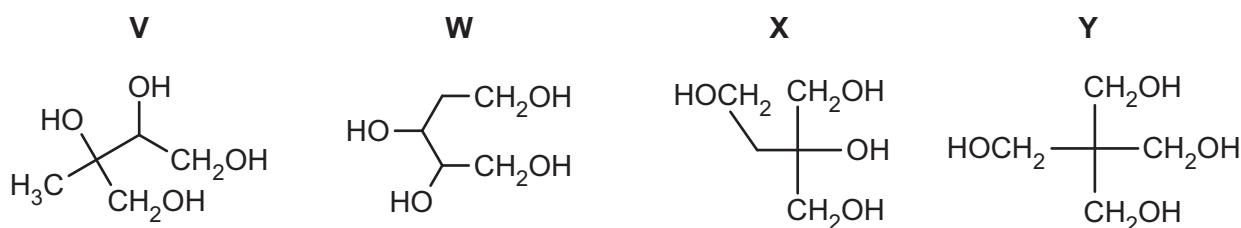
**Figure 4.1**

Context: Fig. 4.1

**Figure 4.1**

Context: part (c) (introduction)

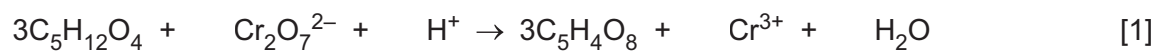
- (c)** Compounds **W**, **X** and **Y** are structural isomers of **V**.

**Figure 4.2**

Context: part (d) (introduction)

- (d)** Exactly 0.03 mol of one of the isomers, **V**, **W**, **X** or **Y**, is completely oxidised by 0.08 mol of acidified  $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$  to produce only **one** organic compound with molecular formula  $\text{C}_5\text{H}_4\text{O}_8$ .

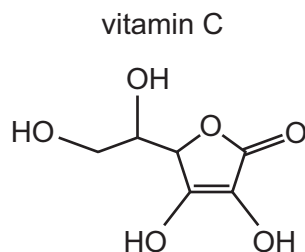
- (ii)** Complete the equation for this reaction.



**5. M/J 25/P21/Q4(b)(iii)**

## Context

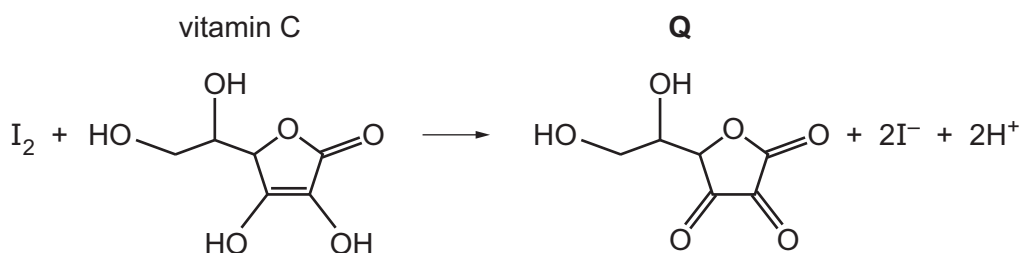
**4** The structure of vitamin C is shown in Fig. 4.1.



**Fig. 4.1**

Context: part (b) (introduction)

**(b)** The concentration of vitamin C is found by titration with  $\text{I}_2(\text{aq})$ .



**Fig. 4.2**

A vitamin C tablet is dissolved in water to produce  $200.0\text{ cm}^3$  of vitamin C solution.  $5.00\text{ cm}^3$  of this vitamin C solution is added to a flask with approximately  $150\text{ cm}^3$  of water and an indicator. Exactly  $28.40\text{ cm}^3$  of  $5.00 \times 10^{-4}\text{ mol dm}^{-3}\text{ I}_2(\text{aq})$  reacts with the sample of vitamin C solution in the flask.

[ $M_r$ : vitamin C, 176]

**(iii)** Deduce the role of  $\text{I}_2(\text{aq})$  in the reaction in Fig. 4.2.

[1]

**6. M/J 25/P24/Q2(b)(ii)**

Context: part (b) (introduction)

(b)  $\text{SCl}_2$  is a red liquid that reacts with water.



(ii) When  $\text{SCl}_2$  reacts with water, the  $\text{SCl}_2$  is broken down and a disproportionation reaction occurs.

State the oxidation numbers of sulfur in  $\text{SCl}_2$  and  $\text{H}_2\text{SO}_3$ .

oxidation number of S in  $\text{SCl}_2$

oxidation number of S in  $\text{H}_2\text{SO}_3$

[2]

**7. MAR 25/P22/Q1(a)**

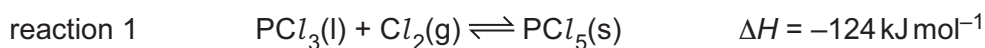
also Unit 7 (2 marks), Unit 2 (2 marks), Unit 4 (2 marks); this unit 2 marks

## Context

**1** Phosphorus and chlorine are elements in Period 3 of the Periodic Table.**(a)** Chlorine forms three different compounds with phosphorus.The most common compounds are  $\text{PCl}_3$  and  $\text{PCl}_5$ .**(i)** Complete Table 1.1.**Table 1.1**

| compound       | oxidation number of P | oxidation number of Cl |
|----------------|-----------------------|------------------------|
| $\text{PCl}_3$ |                       |                        |
| $\text{PCl}_5$ |                       |                        |

[2]

**(ii)** In a closed system,  $\text{PCl}_3$  and  $\text{PCl}_5$  exist in an equilibrium mixture as shown in reaction 1.Deduce **two** conditions that favour the production of  $\text{PCl}_5$  in reaction 1.

1

2

[2]

**(iii)** The third compound of phosphorus and chlorine, **W**, has a relative molecular mass,  $M_r$ , between that of  $\text{PCl}_3$  and  $\text{PCl}_5$ . The compound contains 69.6% by mass of chlorine.Determine the molecular formula of **W**.molecular formula of **W** = [2]**(iv)** **W** is a liquid at room temperature and pressure. It reacts vigorously with water to form an acidic solution.Suggest the structure and bonding in **W**. Explain your answer.

[2]

**8. O/N 24/P21/Q2(c)(ii)-(iii)**

Context

**2** Chlorine is one of the elements in Group 17 of the Periodic Table.

Context: part (c) (introduction)

**(c)**  $\text{CHCl}_3$  is another halogenoalkane.  $\text{CHCl}_3$  forms when propanone reacts with  $\text{NaClO}$ . $\text{NaClO}$  is made from chlorine in a disproportionation reaction.**(ii)** Define disproportionation.

[1]

**(iii)** Write numbers in the boxes to balance the equation showing the reaction of propanone with  $\text{NaClO}$ .

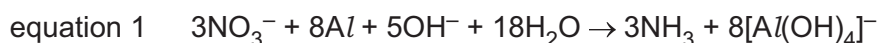
[1]

**9. O/N 24/P22/Q3(b)(i)-(ii)**

Context

**3** Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

Context: part (b) (introduction)

**(b)** A common test for nitrates is the reaction with  $\text{NaOH}$  and  $\text{Al}$ . Equation 1 shows the reaction.**(i)** Deduce the oxidation state of nitrogen in  $\text{NO}_3^-$ .

[1]

**(ii)** Identify the species that is oxidised in equation 1.

[1]

**10. O/N 24/P23/Q2(c)(ii)-(iii)**

Context

**2** Chlorine is one of the elements in Group 17 of the Periodic Table.

Context: part (c) (introduction)

**(c)**  $\text{CHCl}_3$  is another halogenoalkane.  $\text{CHCl}_3$  forms when propanone reacts with  $\text{NaClO}$ . $\text{NaClO}$  is made from chlorine in a disproportionation reaction.**(ii)** Define disproportionation.

[1]

**(iii)** Write numbers in the boxes to balance the equation showing the reaction of propanone with  $\text{NaClO}$ .

[1]

**11. M/J 24/P23/Q1(d)(i)**

Context: part (d) (introduction)

- (d) The oxide of silicon reacts with calcium oxide in an addition reaction to produce calcium silicate,  $\text{CaSiO}_3$ . The oxidation number of calcium in  $\text{CaSiO}_3$  is +II.

- (i) Deduce the oxidation number of silicon in calcium silicate.

oxidation number of silicon

[1]

**12. MAR 24/P22/Q1(b)(i)**

Context

- 1 Bismuth is an element in Group 15 of the Periodic Table.

Context: part (b) (introduction)

- (b) Bismuth reduces water to form bismuth oxide,  $\text{Bi}_2\text{O}_3$ . A colourless gas that ignites with a squeaky pop also forms.

- (i) Construct an equation for the reduction of water by bismuth.

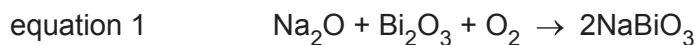
[1]

**13. MAR 24/P22/Q1(c)**

Context

- 1 Bismuth is an element in Group 15 of the Periodic Table.

- (c)  $\text{Bi}_2\text{O}_3$  can be used to form  $\text{NaBiO}_3$ , as shown in equation 1.



- (i) Deduce the oxidation number of Bi in  $\text{Bi}_2\text{O}_3$  and in  $\text{NaBiO}_3$ .

oxidation number of Bi:

in  $\text{Bi}_2\text{O}_3$ in  $\text{NaBiO}_3$ 

[1]

- (ii) Identify the reducing agent in equation 1.

[1]

**14. MAR 24/P22/Q1(d)(i)**

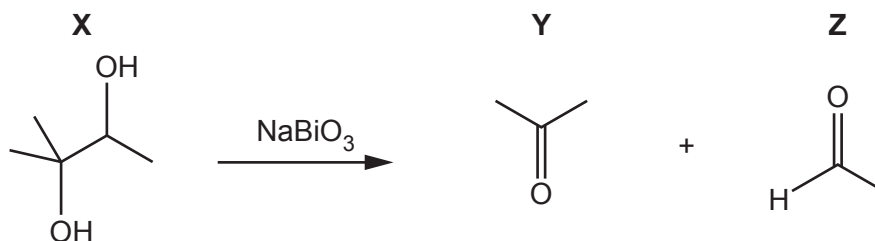
Context

- 1** Bismuth is an element in Group 15 of the Periodic Table.

Context: part (d) (introduction)

- (d)**  $\text{NaBiO}_3$  is an oxidising agent with similar properties to  $\text{KMnO}_4$ .

Fig. 1.1 shows an example of the use of  $\text{NaBiO}_3$  as an oxidising agent.



**Fig. 1.1**

- (i)** Explain the term oxidising agent.

[1]

**15. O/N 23/P21/Q3(a)**

Context

- 3** Potassium chlorate,  $\text{KClO}_3$ , is widely used as an oxidising agent and to make  $\text{O}_2(\text{g})$ .

- (a)** Define oxidising agent.

[1]

**16. O/N 23/P21/Q3(c)**

Context

- 3** Potassium chlorate,  $\text{KClO}_3$ , is widely used as an oxidising agent and to make  $\text{O}_2(\text{g})$ .

- (c)** When  $\text{KClO}_3$  is heated without a catalyst,  $\text{KClO}_4$  and  $\text{KCl}$  form.



Explain why this reaction is described as a disproportionation reaction.

[1]

**17. O/N 23/P22/Q3(b)(iii)**

Context

**3** Phosphoric(V) acid,  $\text{H}_3\text{PO}_4$ , is used in both inorganic and organic reactions.

Context: part (b) (introduction)

**(b)**  $\text{H}_3\text{PO}_4$  is also formed in the process shown in reaction 1.

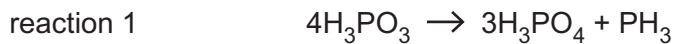


Table 3.1 shows some relevant thermodynamic data.

**Table 3.1**

| compound                | enthalpy change of formation,<br>$\Delta H_f / \text{kJ mol}^{-1}$ |
|-------------------------|--------------------------------------------------------------------|
| $\text{H}_3\text{PO}_3$ | -972                                                               |
| $\text{H}_3\text{PO}_4$ | -1281                                                              |
| $\text{PH}_3$           | +9                                                                 |

**(iii)** Explain why reaction 1 is a disproportionation reaction.

Explain your reasoning with reference to relevant oxidation numbers.

[2]

**18. O/N 23/P23/Q3(a)**

Context

**3** Potassium chlorate,  $\text{KClO}_3$ , is widely used as an oxidising agent and to make  $\text{O}_2(\text{g})$ .

**(a)** Define oxidising agent.

[1]

**19. O/N 23/P23/Q3(c)**

Context

**3** Potassium chlorate,  $\text{KClO}_3$ , is widely used as an oxidising agent and to make  $\text{O}_2(\text{g})$ .

**(c)** When  $\text{KClO}_3$  is heated without a catalyst,  $\text{KClO}_4$  and  $\text{KCl}$  form.



Explain why this reaction is described as a disproportionation reaction.

[1]

**20. M/J 23/P22/Q2(e)(ii)**

Context

**2** Chlorine is a reactive element. It forms many compounds.

Context: part (e) (introduction)

- (e) Bleach is used as a cleaning product to kill bacteria. It is made by adding compounds like sodium chlorate(I),  $\text{NaClO}$ , to water.
- (ii) Sodium chlorate(I),  $\text{NaClO}$ , reacts with hydrogen peroxide to produce sodium chloride, water and oxygen gas.

Construct an equation for this reaction.

[1]

**21. M/J 23/P23/Q1(d)(i)-(ii)**

Context

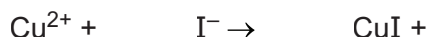
**1** Copper is used in electrical equipment. It has a melting point of  $1085^\circ\text{C}$ .

Context: part (d) (introduction)

- (d) When  $\text{KI(aq)}$  is added to  $\text{CuSO}_4\text{(aq)}$  the blue-coloured solution turns brown and a white precipitate of  $\text{CuI(s)}$  is seen.

The reaction between copper ions and iodide forms only two products.

- (i) Complete the equation for this reaction.



[1]

- (ii) Identify the oxidising agent in this reaction. Explain your answer in terms of electron transfer.

[1]

**22. M/J 22/P22/Q1(b)(i)-(ii)**

Context: part (b) (introduction)

- (b) When magnesium is heated in air, magnesium oxide,  $\text{MgO}$ , is the major product. Smaller amounts of magnesium nitride,  $\text{Mg}_3\text{N}_2$ , are also made.

- (i) Calculate the oxidation number for magnesium and for the nitrogen species in  $\text{Mg}_3\text{N}_2$  to complete Table 1.1.

**Table 1.1**

| species          | magnesium in $\text{Mg}_3\text{N}_2$ | nitrogen in $\text{Mg}_3\text{N}_2$ |
|------------------|--------------------------------------|-------------------------------------|
| oxidation number |                                      |                                     |

[1]

- (ii) Identify the type of reaction which takes place between magnesium and nitrogen. Explain your answer.

[1]

**23. MAR 22/P22/Q2(a)(i)**

Context

**2** Some oxides of elements in Period 3 are shown.



Context: part (a) (introduction)

**(a)** Na reacts with  $\text{O}_2$  to form  $\text{Na}_2\text{O}$ . Na is the reducing agent in this reaction.

**(i)** Define reducing agent.

[1]

**24. MAR 22/P22/Q2(c)(ii)**

Context

**2** Some oxides of elements in Period 3 are shown.



Context: part (c) (introduction)

**(c)**  $\text{P}_4\text{O}_6$  is a white solid that has a melting point of  $24^\circ\text{C}$ . Solid  $\text{P}_4\text{O}_6$  reacts with water to form  $\text{H}_3\text{PO}_3$ .

**(ii)** Determine the oxidation number of P in  $\text{H}_3\text{PO}_3$ .

[1]

## Answers Section

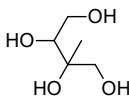
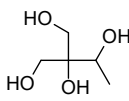
### 1. M/J 26/P21/Q3(d)

|      |                                |                                   |                                                                                                                                                                                                                                        |
|------|--------------------------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3(d) | <b>species</b>                 | <b>oxidation number of sulfur</b> | <b>2</b> •✓✓<br><br><b>A</b> 6 (+) 4 (+) and 2–<br><b>A</b> VI (+) <b>AND</b> IV (+) <b>AND</b> II –<br><br><b>R</b> S <sup>+4</sup> S <sup>+6</sup> or S <sup>–2</sup><br><b>R</b> S <sup>4+</sup> S <sup>6+</sup> or S <sup>2–</sup> |
|      | H <sub>2</sub> SO <sub>4</sub> | (+)VI <b>OR</b> (+)6              |                                                                                                                                                                                                                                        |
|      | SO <sub>2</sub>                | (+)IV <b>OR</b> (+)4              |                                                                                                                                                                                                                                        |
|      | H <sub>2</sub> S               | –II <b>OR</b> –2                  |                                                                                                                                                                                                                                        |

### 2. M/J 26/P21/Q4(a)(ii)

|          |                                                                                                                                                                      |          |  |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--|
| 4(a)(ii) | $\text{C}_5\text{H}_{12}\text{O}_4 + 2\text{Cr}_2\text{O}_7^{2-} + 16\text{H}^+ \rightarrow \text{C}_5\text{H}_6\text{O}_7 + 4\text{Cr}^{3+} + 11\text{H}_2\text{O}$ | <b>1</b> |  |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--|

### 3. M/J 26/P22/Q4(c)

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |                                                                                                                                    |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------|
| 4(c)(i)  | $3\text{C}_5\text{H}_{12}\text{O}_4 + 5\text{Cr}_2\text{O}_7^{2-} + 40\text{H}^+ \rightarrow 3\text{C}_5\text{H}_6\text{O}_8 + 10\text{Cr}^{3+} + 29\text{H}_2\text{O}$<br><br><b>M1</b> $5\text{Cr}_2\text{O}_7^{2-}$<br><b>M2</b> fully correct equation                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>2</b> |                                                                                                                                    |
| 4(c)(ii) | HOCH <sub>2</sub> CH(OH)C(OH)(CH <sub>3</sub> )CH <sub>2</sub> OH <b>OR</b> (HOCH <sub>2</sub> ) <sub>2</sub> C(OH)(CHOHCH <sub>3</sub> )<br><br><div style="display: flex; align-items: center; justify-content: space-around;">  <div><b>OR</b></div>  </div> <ul style="list-style-type: none"> <li>• molecular formula C<sub>5</sub>H<sub>12</sub>O<sub>4</sub></li> <li>• 2 primary alcohol groups</li> <li>• 1 secondary alcohol group</li> <li>• 1 tertiary alcohol group</li> </ul> <div style="text-align: right;">•✓•✓</div> | <b>2</b> | <b>A</b> unambiguous structure<br><b>I</b> any name attempted<br><br>Max 1 all correct but 1 or more connectivity issue e.g. C–H–O |

### 4. M/J 26/P23/Q4(d)(ii)

|          |                                                                                                                                                                         |          |  |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--|
| 4(d)(ii) | $3\text{C}_5\text{H}_{12}\text{O}_4 + 8\text{Cr}_2\text{O}_7^{2-} + 64\text{H}^+ \rightarrow 3\text{C}_5\text{H}_4\text{O}_8 + 16\text{Cr}^{3+} + 44\text{H}_2\text{O}$ | <b>1</b> |  |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--|

### 5. M/J 25/P21/Q4(b)(iii)

|           |                 |          |  |
|-----------|-----------------|----------|--|
| 4(b)(iii) | oxidising agent | <b>1</b> |  |
|-----------|-----------------|----------|--|

### 6. M/J 25/P24/Q2(b)(ii)

|          |                                                                                                                                         |          |  |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------|----------|--|
| 2(b)(ii) | oxidation number of S in $\text{SCl}_2 = (+)2 / (+)\text{II}$<br>oxidation number of S in $\text{H}_2\text{SO}_3 = (+)4 / (+)\text{IV}$ | <b>2</b> |  |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------|----------|--|

### 7. MAR 25/P22/Q1(a)

|           |                                                                                                                                               |          |  |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------|--|
| 1(a)(i)   | <ul style="list-style-type: none"> <li>• (+)3 / III      • –1 / I</li> <li>• (+)5 / V      • –1 / I</li> </ul>                                | <b>2</b> |  |
| 1(a)(ii)  | high pressure<br><br>low temperature                                                                                                          | <b>2</b> |  |
| 1(a)(iii) | ratio of moles of P : Cl = $\frac{30.4}{31.0} : \frac{69.6}{35.5} = 0.98 : 1.96 (= 1 : 2)$<br><br>molecular formula = $\text{P}_2\text{Cl}_4$ | <b>2</b> |  |
| 1(a)(iv)  | (structure =) simple / molecular because low melting point / weak intermolecular forces<br><br>(bonding =) covalent because hydrolysed        | <b>2</b> |  |

**8. O/N 24/P21/Q2(c)(ii)-(iii)**

|           |                                                                                                                 |          |
|-----------|-----------------------------------------------------------------------------------------------------------------|----------|
| 2(c)(ii)  | a reaction where a species / same species is both oxidised and reduced simultaneously / in the same reaction    | <b>1</b> |
| 2(c)(iii) | $\text{CH}_3\text{COCH}_3 + 3\text{NaClO} \rightarrow 1\text{CHCl}_3 + 1\text{CH}_3\text{COONa} + 2\text{NaOH}$ | <b>1</b> |

**9. O/N 24/P22/Q3(b)(i)-(ii)**

|          |                |          |
|----------|----------------|----------|
| 3(b)(i)  | (+)5 / V       | <b>1</b> |
| 3(b)(ii) | aluminium / Al | <b>1</b> |

**10. O/N 24/P23/Q2(c)(ii)-(iii)**

|           |                                                                                                                 |          |
|-----------|-----------------------------------------------------------------------------------------------------------------|----------|
| 2(c)(ii)  | a reaction where a species / same species is both oxidised and reduced simultaneously / in the same reaction    | <b>1</b> |
| 2(c)(iii) | $\text{CH}_3\text{COCH}_3 + 3\text{NaClO} \rightarrow 1\text{CHCl}_3 + 1\text{CH}_3\text{COONa} + 2\text{NaOH}$ | <b>1</b> |

**11. M/J 24/P23/Q1(d)(i)**

|         |       |          |
|---------|-------|----------|
| 1(d)(i) | (+)IV | <b>1</b> |
|---------|-------|----------|

**12. MAR 24/P22/Q1(b)(i)**

|         |                                                                                    |          |
|---------|------------------------------------------------------------------------------------|----------|
| 1(b)(i) | $2\text{Bi} + 3\text{H}_2\text{O} \rightarrow \text{Bi}_2\text{O}_3 + 3\text{H}_2$ | <b>1</b> |
|---------|------------------------------------------------------------------------------------|----------|

**13. MAR 24/P22/Q1(c)**

|          |                                                                  |          |
|----------|------------------------------------------------------------------|----------|
| 1(c)(i)  | $\text{Bi}_2\text{O}_3$ (+)3 / III AND $\text{NaBiO}_3$ (+)5 / V | <b>1</b> |
| 1(c)(ii) | $\text{Bi}_2\text{O}_3$                                          | <b>1</b> |

**14. MAR 24/P22/Q1(d)(i)**

|         |                                                                                          |          |
|---------|------------------------------------------------------------------------------------------|----------|
| 1(d)(i) | a substance that removes electrons from another / causes electron loss / gains electrons | <b>1</b> |
|---------|------------------------------------------------------------------------------------------|----------|

**15. O/N 23/P21/Q3(a)**

|      |                                                                    |          |
|------|--------------------------------------------------------------------|----------|
| 3(a) | compound / molecule / substance that oxidises another / is reduced | <b>1</b> |
|------|--------------------------------------------------------------------|----------|

**16. O/N 23/P21/Q3(c)**

|      |                                            |          |
|------|--------------------------------------------|----------|
| 3(c) | Cl / chlorine is both oxidised and reduced | <b>1</b> |
|------|--------------------------------------------|----------|

**17. O/N 23/P22/Q3(b)(iii)**

|           |                                                                                         |          |
|-----------|-----------------------------------------------------------------------------------------|----------|
| 3(b)(iii) | P (in $\text{H}_3\text{PO}_3$ ) is (both) oxidised and reduced (simultaneously)         | <b>1</b> |
|           | P (is oxidised) from (+)3 / (+)III $\rightarrow$ (+)5 / (+)V AND (reduced to) -3 / -III | <b>1</b> |

**18. O/N 23/P23/Q3(a)**

|      |                                                                    |          |
|------|--------------------------------------------------------------------|----------|
| 3(a) | compound / molecule / substance that oxidises another / is reduced | <b>1</b> |
|------|--------------------------------------------------------------------|----------|

**19. O/N 23/P23/Q3(c)**

|      |                                            |          |
|------|--------------------------------------------|----------|
| 3(c) | Cl / chlorine is both oxidised and reduced | <b>1</b> |
|------|--------------------------------------------|----------|

**20. M/J 23/P22/Q2(e)(ii)**

|          |                                                                                                 |          |
|----------|-------------------------------------------------------------------------------------------------|----------|
| 2(e)(ii) | $\text{NaClO} + \text{H}_2\text{O}_2 \rightarrow \text{O}_2 + \text{NaCl} + \text{H}_2\text{O}$ | <b>1</b> |
|----------|-------------------------------------------------------------------------------------------------|----------|

**21. M/J 23/P23/Q1(d)(i)-(ii)**

|          |                                                                                                                        |          |
|----------|------------------------------------------------------------------------------------------------------------------------|----------|
| 1(d)(i)  | $2\text{Cu}^{2+} + 4\text{I}^{-} \rightarrow 2\text{CuI} + \text{I}_2$                                                 | <b>1</b> |
| 1(d)(ii) | $\text{Cu}^{2+}$ OR $\text{CuSO}_4$ <b>AND</b> copper (species / ion) has gained / taken electron(s) from iodide (ion) | <b>1</b> |

**22. M/J 22/P22/Q1(b)(i)-(ii)**

|          |                                                                                                                                                                                       |                                      |                                     |          |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|-------------------------------------|----------|
| 1(b)(i)  | species                                                                                                                                                                               | magnesium in $\text{Mg}_3\text{N}_2$ | nitrogen in $\text{Mg}_3\text{N}_2$ | <b>1</b> |
|          | oxidation number                                                                                                                                                                      | 2(+) / (+)II                         | 3- / -III                           |          |
| 1(b)(ii) | <ul style="list-style-type: none"> <li>redox</li> <li>Mg loses electrons / Mg increases ox no / Mg is oxidised</li> <li>N gains electrons / N reduces ox no / N is reduced</li> </ul> |                                      |                                     | <b>1</b> |

**23. MAR 22/P22/Q2(a)(i)**

|         |                                |          |
|---------|--------------------------------|----------|
| 2(a)(i) | species that donates electrons | <b>1</b> |
|---------|--------------------------------|----------|

**24. MAR 22/P22/Q2(c)(ii)**

|          |            |          |
|----------|------------|----------|
| 2(c)(ii) | (+)3 / III | <b>1</b> |
|----------|------------|----------|